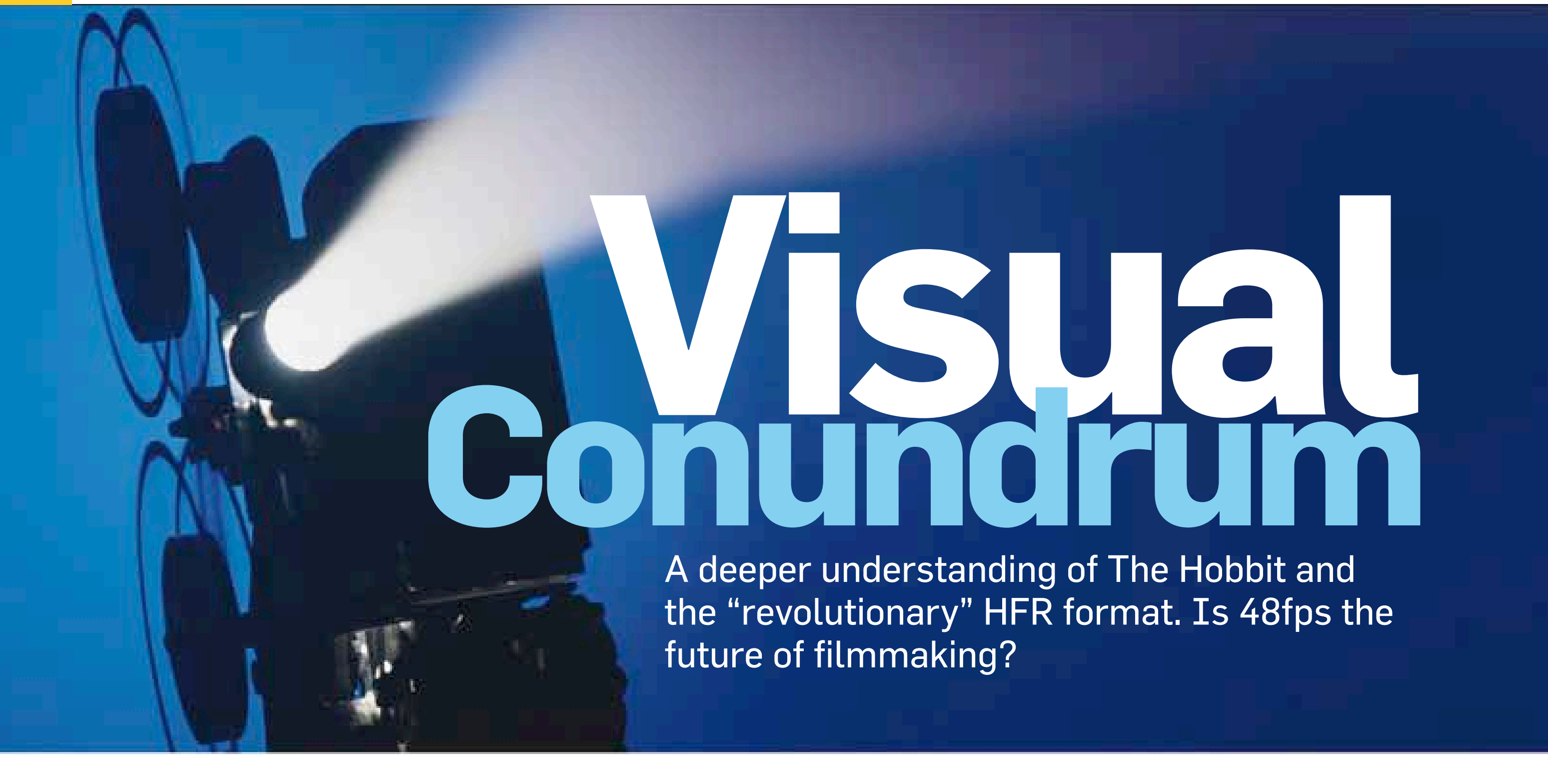
Spiked robots to explore space?
Researchers from NASA and Stanford think that spherical robots with "spikes" are the best way to explore space
http://dgit.in/Up3oRJ

Microsoft buys
Keeping the Xbox in mind, Microsoft has acquired a small home-entertainment startup called "R2 Studios"



Visual Conundrum

A deeper understanding of The Hobbit and the "revolutionary" HFR format. Is 48fps the future of filmmaking?

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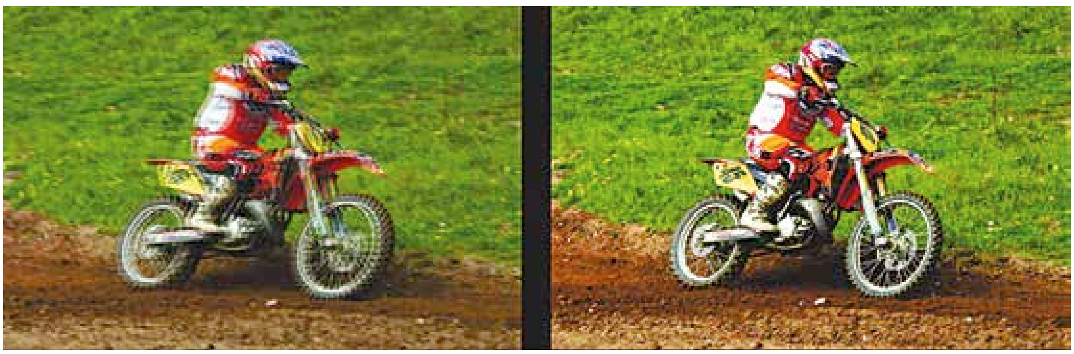
In December of 2012 a new movie was released, a movie that was said to implement a technology that would change the face of movies forever. The movie was *The Hobbit: An Unexpected Journey*. The technology – 48 fps HFR or High-Frame-Rate as it is better known. Whether the movie lived up to its expectations is a matter of debate but what about HFR?

Movie vs. Video

To better understand how HFR affects us, as an audience, we need to understand the difference between a video and a movie and yes, there is a very big difference between the two. A video is nothing more than a collection of images that are displayed in quick succession, giving the illusion of motion, usually accompanied by audio. A movie is a video that is supposed to be a work of art. Someone's vision that has been painstakingly brought to life. A good movie tells a tale, transports you to another world and gets you completely involved. A bad movie is just a video that attempts to do all of the above.

What matters, or should matter to the director, is how the movie is experienced by the people, the audience. The choice of frame-rate is as important to the scene and the movie as a whole as is the choice of cast and crew, colour or monochrome. Movies are normally shot at a rate of 24 frames per second (fps) and this value was standardised for various reasons, but primarily because this was the minimum frame-rate at which our brain can process still images and be fooled into thinking it was looking at motion. The key word here is "minimum". While adequate, this frame-rate is far from ideal and was chosen more for the fact that film (think negatives, not movies) was rather expensive in those days and this was the cheapest frame-rate required to produce a movie, than for the fact that this was an ideal fps to showcase a movie in. The 24 fps standard is actually very prone to flickering (hence the name, "flicks" associated with movies) and today's projectors have to compensate by projecting at least two duplicate frames per frame projected to compensate for this flickering. As an audience, we adapted to this frame-rate, it was double that of the 12 fps from the



Blurred? 24 fps. Crystal clear? 28 fps. Which is better depends entirely on the context in which the video is shown. Good enough for a movie, bad for sports.